

5.4 Least squares and the normal equations

The straight-line calculation of [Section 5.2](#) extends to any model matrix. The algebra is shorter in matrix form than it was for two coefficients, because the derivatives of linear and quadratic forms were worked out once and for all in [Chapter 1](#). In this section $\mathbf{y}$ is a fixed data vector. Nothing here uses probability. The statistical properties come in [Section 5.5](#).

5.4.1 The least squares estimate

Definition 5.4.1 — Least squares estimate

For an $n \times p$ matrix $\mathbf{X}$ and $\mathbf{y} \in \mathbb{R}^n$, the **least squares criterion** is

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \mathbf{b} \in \mathbb{R}^p.$$

A **least squares estimate** of β is any $\hat{\beta}$ that minimizes S . The corresponding **fitted values** are $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$, the **residuals** are $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\beta}$, and the **residual sum of squares** is $\text{SSE} = S(\hat{\beta}) = \|\hat{\boldsymbol{\varepsilon}}\|^2$.

Following the book's notation, we write $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ in bold for the vectors, and $\hat{y}_i$ and $\hat{\varepsilon}_i$ for their entries, whether the data are regarded as fixed numbers or as random.

Theorem 5.4.1 — Least squares for a full-rank model

Let $\mathbf{X}$ be an $n \times p$ matrix of rank p . Then:

- (a) $\mathbf{X}^\top \mathbf{X}$ is positive definite, and in particular nonsingular;
- (b) the **normal equations**

$$\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y} \tag{5.4.1}$$

have exactly one solution,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}; \tag{5.4.2}$$

- (c) for every $\mathbf{b} \in \mathbb{R}^p$,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 + (\mathbf{b} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X}(\mathbf{b} - \hat{\beta}), \tag{5.4.3}$$

so $\hat{\beta}$ is the unique least squares estimate.

If instead $\text{rank}(\mathbf{X}) < p$, least squares estimates exist but are never unique.

Proof. (a) $\mathbf{X}^\top \mathbf{X}$ is symmetric, and $\mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b} = \|\mathbf{X}\mathbf{b}\|^2 \geq 0$, with equality iff $\mathbf{X}\mathbf{b} = \mathbf{0}$. Since $\mathbf{X}$ has rank p , rank-nullity (Theorem 1.2.2) gives $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, so the quadratic form is positive for $\mathbf{b} \neq \mathbf{0}$. A positive definite matrix is nonsingular (Proposition 1.7.5(e)).

(b) This is the computation of Example (The least squares criterion): by Theorem 1.11.1 the gradient of S is $-2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b})$ and its Hessian is $2\mathbf{X}^\top\mathbf{X}$. Every minimizer of the differentiable function S is a stationary point, and the stationary points are the solutions of Eq. 5.4.1. By (a) there is exactly one, Eq. 5.4.2.

(c) Put $\mathbf{d} = \mathbf{b} - \hat{\boldsymbol{\beta}}$. Then $\mathbf{y} - \mathbf{X}\mathbf{b} = \hat{\boldsymbol{\varepsilon}} - \mathbf{X}\mathbf{d}$, and

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 - 2\mathbf{d}^\top\mathbf{X}^\top\hat{\boldsymbol{\varepsilon}} + \mathbf{d}^\top\mathbf{X}^\top\mathbf{X}\mathbf{d}.$$

The middle term vanishes because $\mathbf{X}^\top\hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top\mathbf{y} - \mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}$ by the normal equations. The last term is positive for $\mathbf{d} \neq \mathbf{0}$ by (a). So $S(\mathbf{b}) > S(\hat{\boldsymbol{\beta}})$ whenever $\mathbf{b} \neq \hat{\boldsymbol{\beta}}$.

For the last statement, suppose $\text{rank}(\mathbf{X}) < p$. The vector $\mathbf{X}^\top\mathbf{y}$ lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top\mathbf{X})$ (Proposition 1.2.3(c)), so by Proposition 1.11.2, applied with $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ and $S(\mathbf{b}) = \mathbf{y}^\top\mathbf{y} + (\mathbf{b}^\top\mathbf{X}^\top\mathbf{X}\mathbf{b} - 2\mathbf{b}^\top\mathbf{X}^\top\mathbf{y})$, the minimizers of S are exactly the solutions of the normal equations, and they exist. But there is a $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$, and then $S(\mathbf{b} + t\mathbf{v}) = S(\mathbf{b})$ for every t , so no minimizer is unique.

■

The calculus proof finds the stationary point and then needs a separate argument, Eq. 5.4.3, to show it is the global minimum. The Hessian $2\mathbf{X}^\top\mathbf{X}$ tells the same story: it is positive definite, so S is a strictly convex quadratic bowl with a single lowest point. The name “normal equations” has nothing to do with the normal distribution. It refers to the orthogonality they express: $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ says the residual vector is *normal*, that is, perpendicular, to every column of $\mathbf{X}$. Chapter 6 makes that geometry the starting point, derives the normal equations again without calculus (Theorem 6.4.2), and treats the rank-deficient case in full.

Remark (The rank-deficient case). When $\text{rank}(\mathbf{X}) < p$, the proof shows that the least squares estimates form an affine set $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X})$ of dimension $p - \text{rank}(\mathbf{X})$. Different programs report different members of this set, according to which columns they drop or which constraints they impose. The fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ are the same for all of them, since $\mathbf{X}(\hat{\boldsymbol{\beta}} + \mathbf{v}) = \mathbf{X}\hat{\boldsymbol{\beta}}$ for $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. Which linear combinations of the coefficients share this invariance is the question of estimability, answered in Chapter 8 (Definition 8.2.1 and Theorem 8.3.1).

Least squares under known linear restrictions $\mathbf{H}\mathbf{b} = \mathbf{h}$, such as shares that add to one, has a closed form found by Lagrange multipliers in Exercise (B1); the increase in the residual sum of squares it computes drives the test of the restrictions in Chapter 11 (Theorem 11.3.2).

Example. (The straight line again)

For $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$,

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}, \quad \mathbf{X}^\top\mathbf{y} = \begin{pmatrix} n\bar{y} \\ \sum_i x_i y_i \end{pmatrix}.$$

The determinant is $n \sum_i x_i^2 - n^2 \bar{x}^2 = nS_{xx}$, positive iff the x_i are not all equal, which is exactly when $\text{rank}(\mathbf{X}) = 2$. Inverting,

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{pmatrix}, \quad (5.4.4)$$

using $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$. The second entry of $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is $(\sum_i x_i y_i - n\bar{x}\bar{y})/S_{xx} = S_{xy}/S_{xx}$, and the first is $\bar{y} - \bar{x}S_{xy}/S_{xx}$. This recovers Theorem 5.2.1, and Eq. 5.4.3 reduces to Eq. 5.2.5.

5.4.2 Fitted values, residuals and the hat matrix

Substituting Eq. 5.4.2, the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad (5.4.5)$$

and the residuals are $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$. The $n \times n$ matrix $\mathbf{H}$ is the **hat matrix**, so called because it puts the hat on $\mathbf{y}$. It depends only on the regressors, never on the response.

Proposition 5.4.2 — Algebra of the fit

Let $\mathbf{X}$ have rank p , and let $\mathbf{H}$, $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ be as above.

- (a) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$, and $\text{tr } \mathbf{H} = p$, $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$.
- (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$.
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$, and $\text{SSE} = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$.
- (d) If $\mathbf{1} = \mathbf{X}\mathbf{c}$ for some $\mathbf{c}$ (for example, if $\mathbf{X}$ has an intercept column), then $\sum_i \hat{\varepsilon}_i = 0$, the mean of the fitted values is $\bar{y}$, and

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

that is, $\text{SST} = \text{SSR} + \text{SSE}$.

- (e) If $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$ and $\boldsymbol{\varepsilon}$, then $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$.

Proof. (a) $\mathbf{H}^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{H}$, since the inverse of a symmetric matrix is symmetric. $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$, and then $\mathbf{H}^2 = \mathbf{H}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$. By the cyclic property (Theorem 1.6.1), $\text{tr } \mathbf{H} = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr } \mathbf{I}_p = p$. (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = ((\mathbf{I} - \mathbf{H})\mathbf{X})^\top \mathbf{y} = \mathbf{0}$, and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (c) $\mathbf{y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$ with $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$ gives the first identity. For the second, $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent, so $\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y}$; and $\mathbf{y}^\top \mathbf{H}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$. (d) $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{c}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$ by (b). Then $\sum_i \hat{y}_i = \sum_i y_i$. Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$. The cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i = \hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} - \bar{y} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (e) Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into Eq. 5.4.2 and into $(\mathbf{I} - \mathbf{H})\mathbf{y}$, using (a). ■

Part (e) will carry most of the statistical theory. The estimation error $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ is a fixed linear

function of the unobserved errors, and the residual vector does not depend on β at all. Part (b) says that the residuals satisfy p independent linear constraints. That is the origin of the divisor $n - p$ in [Section 5.6](#).

For a model with an intercept, part (d) motivates the **coefficient of determination**

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

the proportion of the variation of the response about its mean that the fitted values reproduce. It generalizes Proposition 5.2.2(d), and lies in $[0, 1]$. It can only increase when a column is added to $\mathbf{X}$ ([Exercise \(B1\)](#)), which is why [Chapter 9](#) introduces an adjusted version (Proposition 9.5.3). [Section 6.7](#) interprets R as the cosine of an angle.

5.4.3 Centring the regressors

For a model with an intercept, write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, where $\mathbf{X}_1$ is $n \times k$ with column means $\bar{\mathbf{x}} = \mathbf{X}_1^\top \mathbf{1}/n$. The **centred regressors** are the columns of

$$\mathbf{X}_c = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top = \left(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbf{X}_1,$$

each of which sums to zero. The next result is the matrix version of the formulas $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.

Proposition 5.4.3 — Centred form

Let $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ have rank $p = k + 1$, and partition $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1^\top)^\top$. Then $\mathbf{X}_c$ has rank k , and

$$\hat{\beta}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y} = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1}), \quad \hat{\beta}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\beta}_1.$$

Proof. For $\mathbf{b} = (b_0, \mathbf{b}_1^\top)^\top$,

$$\mathbf{X}\mathbf{b} = b_0\mathbf{1} + \mathbf{X}_1\mathbf{b}_1 = (b_0 + \bar{\mathbf{x}}^\top \mathbf{b}_1)\mathbf{1} + \mathbf{X}_c\mathbf{b}_1 = [\mathbf{1}, \mathbf{X}_c]\mathbf{T}\mathbf{b}, \quad \mathbf{T} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}.$$

$\mathbf{T}$ is nonsingular (it is upper triangular with unit diagonal), so $[\mathbf{1}, \mathbf{X}_c] = \mathbf{X}\mathbf{T}^{-1}$ has rank p (Proposition 1.2.3(b)), and hence $\mathbf{X}_c$ has rank k . The criterion satisfies $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - [\mathbf{1}, \mathbf{X}_c]\mathbf{a}\|^2$ with $\mathbf{a} = \mathbf{T}\mathbf{b}$, and $\mathbf{b} \mapsto \mathbf{T}\mathbf{b}$ is a bijection of $\mathbb{R}^p$. So $\hat{\beta}$ minimizes the left side iff $\hat{\mathbf{a}} = \mathbf{T}\hat{\beta}$ minimizes the right side. Because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$, the normal equations for $[\mathbf{1}, \mathbf{X}_c]$ are block diagonal,

$$\begin{pmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{X}_c^\top \mathbf{X}_c \end{pmatrix} \hat{\mathbf{a}} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c^\top \mathbf{y} \end{pmatrix},$$

giving $\hat{a}_0 = \bar{y}$ and $\hat{\mathbf{a}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y}$. Undoing $\mathbf{T}$: $\hat{\beta}_1 = \hat{\mathbf{a}}_1$ and $\hat{\beta}_0 = \hat{a}_0 - \bar{\mathbf{x}}^\top \hat{\mathbf{a}}_1$. Finally $\mathbf{X}_c^\top \mathbf{1} = \mathbf{0}$, so $\mathbf{X}_c^\top \mathbf{y} = \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1})$. ■

In words: to find the slopes, centre everything and fit without an intercept; the intercept then makes the fitted surface pass through the point of means $(\bar{\mathbf{x}}, \bar{y})$. The matrix $\mathbf{X}_c^\top \mathbf{X}_c / (n - 1)$ is the

sample covariance matrix of the regressors, and $\mathbf{X}_c^\top \mathbf{y} / (n - 1)$ their sample covariances with the response. So the slopes are “covariance divided by variance”, as in the straight-line case, with a matrix inverse in place of the division. This is the simplest case of the Frisch–Waugh–Lovell theorem of [Section 6.6](#), with the intercept playing the part of the regressors that are “partialled out”.

5.4.4 Indicator variables

The least squares estimates in a model with only indicator columns are simple averages.

Proposition 5.4.4 — Coefficients of indicator variables

Suppose each case belongs to one of $c \geq 2$ groups, group k has $n_k \geq 1$ cases with mean response $\bar{y}_k$, and $\mathbf{z}_k$ is the indicator column of group k . Let $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_{c-1}]$, so that group c is the reference. Then $\mathbf{X}$ has full rank, and

$$\hat{\beta}_0 = \bar{y}_c, \quad \hat{\beta}_k = \bar{y}_k - \bar{y}_c \quad (k = 1, \dots, c - 1).$$

The fitted value of every case is the mean of its group.

Proof. If $\mathbf{X}\mathbf{b} = \mathbf{0}$, the entries for a case in group c give $b_0 = 0$, and those for a case in group $k < c$ give $b_0 + b_k = 0$, so $\mathbf{b} = \mathbf{0}$: the rank is $p = c$. Let $\mathbf{b}$ be the proposed vector. Its fitted value for a case in group $k < c$ is $\bar{y}_c + (\bar{y}_k - \bar{y}_c) = \bar{y}_k$, and for a case in group c it is $\bar{y}_c$. The residuals $y_i - \bar{y}_{k(i)}$ therefore sum to zero within each group, so $\mathbf{z}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = 0$ for every $k = 1, \dots, c$. Every column of $\mathbf{X}$ is a sum of group indicators, since $\mathbf{1} = \mathbf{z}_1 + \dots + \mathbf{z}_c$, so $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$: the normal equations hold. By [Theorem 5.4.1](#), $\mathbf{b} = \hat{\beta}$. ■

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
X = np.column_stack([np.ones(len(y)), D]) # excellent =
    reference

beta_hat = np.linalg.solve(X.T @ X, X.T @ y)
level = D @ np.array([1, 2, 3]) # 0 = excellent, ...,
    3 = poor
means = np.array([y[level == k].mean() for k in range(4)])
print("coefficients:      ", np.round(beta_hat, 4))
print("group means:      ", np.round(means, 4))
print("means minus reference: ", np.round(means - means[0], 4))
```

[Run this code online](#)

For the health data of Example (Self-rated health) the group sizes are 11,019 (excellent), 7,309 (good), 1,560 (fair) and 302 (poor), and the group means are 2.634, 2.902, 3.692 and 5.795 visits. The listing confirms that the coefficients are these means’ differences from the reference.

5.4.5 A worked example

Example. (Money demand in Denmark)

Economic theory relates the amount of money people choose to hold to their income, which drives the volume of transactions, and to interest rates, which measure the return given up by holding money rather than bonds. A classic data set records, for $n = 55$ quarters from 1974 to 1987 in Denmark, the logarithm of real money holdings (y), the logarithm of real income, the interest rate on bonds and the interest rate on deposits (rates as fractions per year). The model has $p = 4$ columns: an intercept and the three regressors. Solving the normal equations gives

$$\hat{\beta} = (4.3945, 1.2958, -2.6163, 0.6186)^\top,$$

with residual sum of squares $SSE = 0.09253$ and $R^2 = 0.9262$. The listing checks that $\mathbf{X}^\top \hat{\epsilon}$ is zero up to rounding error, that the two formulas for SSE in [Proposition 5.4.2\(c\)](#) agree, and that the slopes computed from centred data ([Proposition 5.4.3](#)) are the same. [Section 5.6](#) attaches standard errors to these numbers, and [Section 5.7](#) interprets them.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
print("n =", n, " p =", p)
```

[Run this code online](#)

```
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

print("beta_hat =", np.round(beta_hat, 4))
print("X^T e      =", X.T @ resid)          # zero up to rounding
print("SSE two ways:", SSE, y @ y - beta_hat @ Xty)
```

[Run this code online](#)

```
Z = X[:, 1:] - X[:, 1:].mean(axis=0)          # centred regressors, no intercept column
slopes = np.linalg.solve(Z.T @ Z, Z.T @ (y - y.mean()))
intercept = y.mean() - X[:, 1:].mean(axis=0) @ slopes
print("slopes from centred data:", np.round(slopes, 4))
print("intercept recovered:      ", round(intercept, 4))
```

[Run this code online](#)

Warning (Formulas are not algorithms). The formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is for reasoning, not for computing. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of the problem, and inverting it wastes work and accuracy. The listings solve the normal equations with a linear solver, which is acceptable for small, well-conditioned problems like these. Good

software computes $\hat{\beta}$ from an orthogonal factorization of $\mathbf{X}$ itself. [Section 6.10](#) shows the loss of accuracy in an example, and [Chapter 10](#) develops the algorithms (Theorem 10.2.1, Theorem 10.3.2).

5.4.6 Exercises

5.4.6.1 A. Check your understanding

Exercise A1.

Suppose the columns of $\mathbf{X}$ are nonzero and mutually orthogonal. Show that $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{y} / \mathbf{x}_j^\top \mathbf{x}_j$ for each j , so that each coefficient is the same as in the regression of $\mathbf{y}$ on $\mathbf{x}_j$ alone. Show also that deleting a column does not change the other estimates.

5.4.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}]$ have full rank, and let SSE_1 and SSE be the residual sums of squares from regressing $\mathbf{y}$ on $\mathbf{X}_1$ and on $\mathbf{X}$. Show that $\text{SSE} \leq \text{SSE}_1$, directly from [Definition 5.4.1](#). Deduce that R^2 cannot decrease when a column is added to a model with an intercept.

Solution. Let $\hat{\beta}_1$ be the least squares estimate for $\mathbf{X}_1$. The vector $(\hat{\beta}_1^\top, 0)^\top$ is one candidate in the minimization over $\mathbf{b} \in \mathbb{R}^p$, and it gives the value $\|\mathbf{y} - \mathbf{X}_1 \hat{\beta}_1\|^2 = \text{SSE}_1$. The minimum over all candidates is at most this. With an intercept in $\mathbf{X}_1$, $R^2 = 1 - \text{SSE}/\text{SST}$ and SST does not depend on the model, so R^2 cannot decrease. Equality holds iff the extra coefficient in the larger fit is zero, because by [Eq. 5.4.3](#) the candidate $(\hat{\beta}_1^\top, 0)^\top$ is optimal only if it equals $\hat{\beta}$.

Exercise B2.

Let $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ with all $w_i > 0$, and $\mathbf{X}$ of full rank. Show that $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ has the unique minimizer $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$, by applying [Theorem 5.4.1](#) to the rescaled data $\mathbf{W}^{1/2} \mathbf{y}$ and $\mathbf{W}^{1/2} \mathbf{X}$. Which cases get most influence when the weights are the reciprocals of the error variances?

Solution. $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 = \|\mathbf{W}^{1/2} \mathbf{y} - \mathbf{W}^{1/2} \mathbf{X} \mathbf{b}\|^2$. The matrix $\mathbf{W}^{1/2} \mathbf{X}$ has rank p , because $\mathbf{W}^{1/2}$ is nonsingular (Proposition 1.2.3(b)). By [Theorem 5.4.1](#) the unique minimizer is $((\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{X})^{-1} (\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{y} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$. With $w_i = 1/\text{Var}(Y_i)$, the least variable cases count most. This is weighted least squares, the subject of Chapter 21 and, with a general covariance matrix, of [Section 6.9](#).

Exercise B3.

Show that $\hat{\beta} = \mathbf{A} \mathbf{y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, that $\mathbf{A} \mathbf{X} = \mathbf{I}_p$, and that $\mathbf{A} \mathbf{A}^\top =$

$(\mathbf{X}^\top \mathbf{X})^{-1}$. Give an example of a different $p \times n$ matrix $\mathbf{B}$ with $\mathbf{B}\mathbf{X} = \mathbf{I}_p$ when $n > p$. ([Chapter 7](#) compares all such estimators $\mathbf{B}\mathbf{y}$.)

5.4.6.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full rank and let a new case $(\mathbf{x}_0, y_0)$ be appended. Using Theorem 1.6.5, show that the new estimate is

$$\hat{\beta}_{\text{new}} = \hat{\beta} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \hat{\beta})}{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

Interpret: the estimate moves in proportion to the new case's prediction error. This is the basis of *recursive least squares*; [Chapter 10](#) treats the numerically stable version.